Impact of the noise-pollution on *Stenella coeruleoalba* in Sicily

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The area surrounding Sicily and the Strait of Sicily is a Mediterranean biodiversity hotspot with high productivity. Its geographical features, banks, seamounts, mud, and volcanic seamounts, create phytoplankton retention regions and sustain a variety of habitat types. These characteristics increase the productivity of the Sicilian continental shelf and its entire area. Unfortunately, in this area the marine ecosystem and biodiversity is extremely affected by a great number of anthropogenic activities. [1].

Cetacean population in the Mediterranean, are currently affected by the complex of threats deriving from a variety of human activities. However, I would like to consider the Mediterranean striped dolphin (*Stenella coeruleoalba*) that has been assessed as "Vulnerable" according to the IUCN Red List of Threatened Species due to a suspected reduction in population of >30% over the past 60 years [2]. Even tough, this species belongs to the cetaceans legally protected by many national and international agreements, it suffers from several threats due to anthropic activities.

One of the conservation problems of this species in the area around Sicily is the impact of the noise-pollution. The total amount of anthropogenic noise released into the marine environment has increased at a remarkable rate over the last 100 years [3]. As a result of the expansion of industrial activities in the marine environment, including commercial shipping, oil and gas exploration and production, commercial fishing and, more recently, the development of offshore renewable energy [4]. The excess or increase of environmental noise, could have a very serious impact on animals that base their biological functions and behaviours (such as foraging, predation, mating, communication, orientation, etc.) on acoustic signals. *Stenella coeruleoalba* are really sensible to the increasing level of ambient noise, and the scientific literature also assesses the relationship between cetacean strandings and noise-pollution [5].

The occurrences of *Stenella coeruleoalba* (using records from 2010 to 2023), in the area around Sicily, are quite spread around the island, as we can see in the MAP1, but they are not so concentrated in any of the Protected Marine Area (Natura2000) and especially in the south of the island. These small cetaceans are found to distribute themselves close to submarine slopes and escarpments of the strait of Sicily [1].

This study focussed on a set of human activities using noise sources identified as being of primary concern for cetacean conservation [5]:

- Marine traffic (as density of all type of vessels, with annual average 2017-2022) (MAP2)
- Harbours and other costal constructions as urban wastewater discharge points and aquaculture of shellfish these two were considered in the study in connection with the fact that they are sources of pollution and can change local ecology where tidal and other flushing is minimal. Also, may compete for limited habitat access with foraging, resting, socializing and nurturant mammals [6,7].) (MAP3)
- Offshore constructions (marine renewable energy projects, hydrocarbon platform operations, wind farms) (MAP3)

Underwater sound travels very efficiently, therefore thousands of square kilometres or more may be affected. Furthermore, as MAP 4 shows, the accumulation of human activities that generate noise can result in behavioural reactions such intense avoidance of the sound source. Intense levels of sound exposure, that can lead to habitat displacement [8]. On the other hand, we can see that exist a clear interaction and conflicts between striped dolphins and all human activities. According to this data, the *Stenella coeruleoalba*, keeps distance from the south of the Sicily, and this can be due to the high noise disturbance created by the offshore constructions and the vessel traffic. Also, the fact that the dolphins are keeping distance from the Protected area represent an outlandish setting, the areas that are supposed to protect them are not so welcoming after all, because coinciding with the main drivers of the noise pollution.

The realisation that mining operations are incompatible with the lives of cetaceans could be a step taken after this study. Eventually, efforts to establish a Marine Protected Area with a more practical approach for realistic protection might be made, and seamounts and/or submarine canyons could be given priority as the primary areas of biodiversity interest.

Spatial data:

- **Number of encountered species in the study area:**

1. GBIF.org (2023) (TSV) GBIF Occurrence Download https://doi.org/10.15468/dl.jtuy39
2. OBIS (2023) [Data Distribution records of Stenella coeruleoalba (Meyer, 1833)] [Dataset] (Available: Ocean Biodiversity Information System. Intergovernmental Oceanographic Commission of UNESCO. (https://datasets.obis.org/downloads/f3daf35d-936c-47ac-a5bd-830ebd1a6589.zip)

- **Human activities:**

1. **Vessel Density**: Cogea, (2019), EMODnet Human Activities, Vessel Density, {GeoTIFF format} EMODnet. Retrieved from https://emodnet.ec.europa.eu/geonetwork/srv/eng/catalog.search#/metadata/0f2f3ff1-30ef-49e1-96e7-8ca78d58a07c
2. **Harbour:** Eurofish and Cogea, (2014), EMODNET Human Activities, Main Ports, {points} EMODnet. Retrieved from https://emodnet.ec.europa.eu/geonetwork/srv/eng/catalog.search#/metadata/379d0425-8924-4a41-a088-1a002d2ea748
3. **Urban waste discharge points:** AZTI, (2021) EMODnet Human Activities, Urban Waste Water, {points} EMODnet. Retrieved from https://emodnet.ec.europa.eu/geonetwork/srv/eng/catalog.search#/metadata/0bd23b5e-b288-4273-b8b7-d073538ada52
4. **Aquaculture of shellfish:** AND-International, (2014) EMODnet Human Activities: Shellfish aquaculture {points} EMODnet. Retrieved from https://emodnet.ec.europa.eu/geonetwork/srv/eng/catalog.search#/metadata/aa0d2b45-49c4-4b42-86bb-8971a3c2d2cc
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